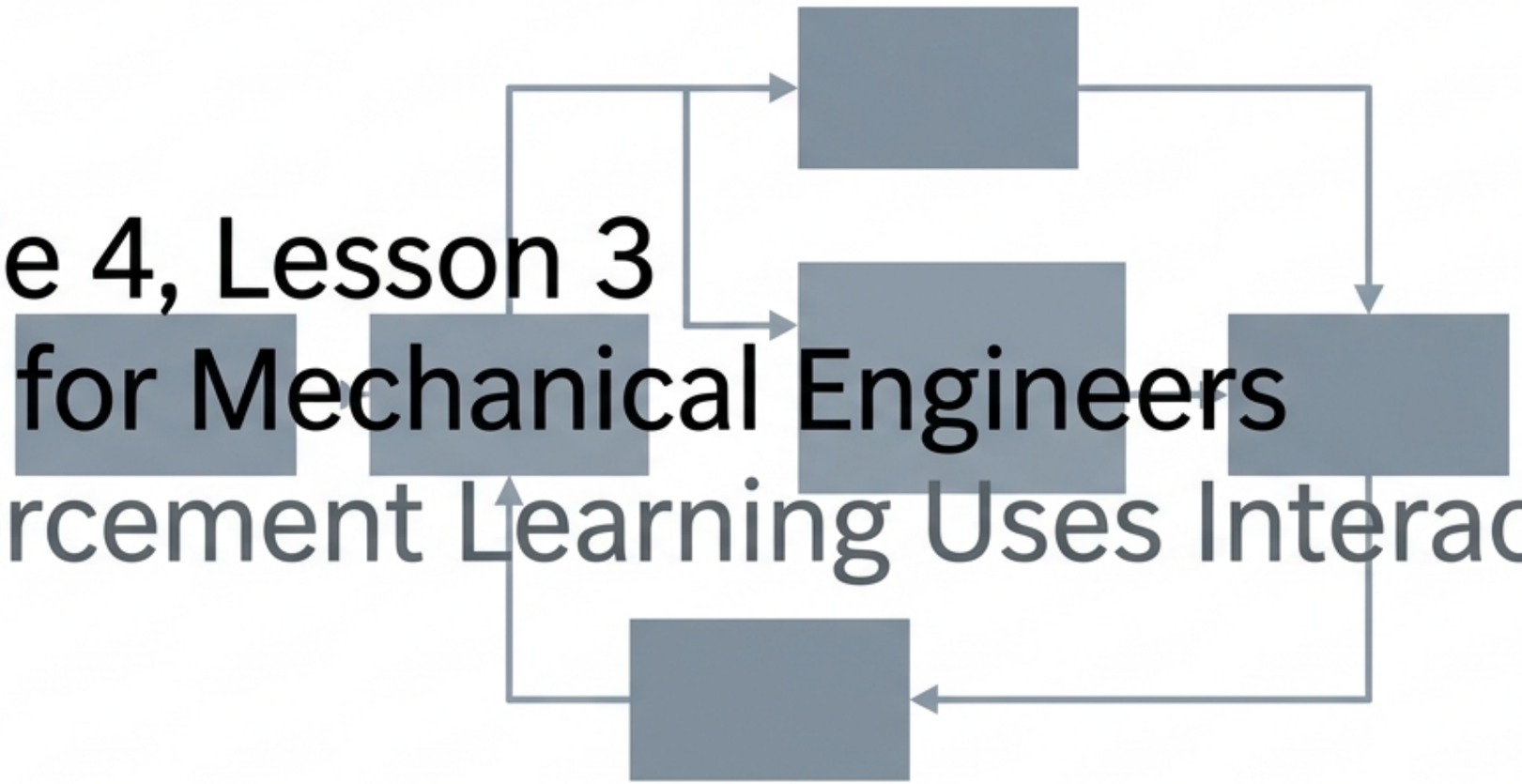
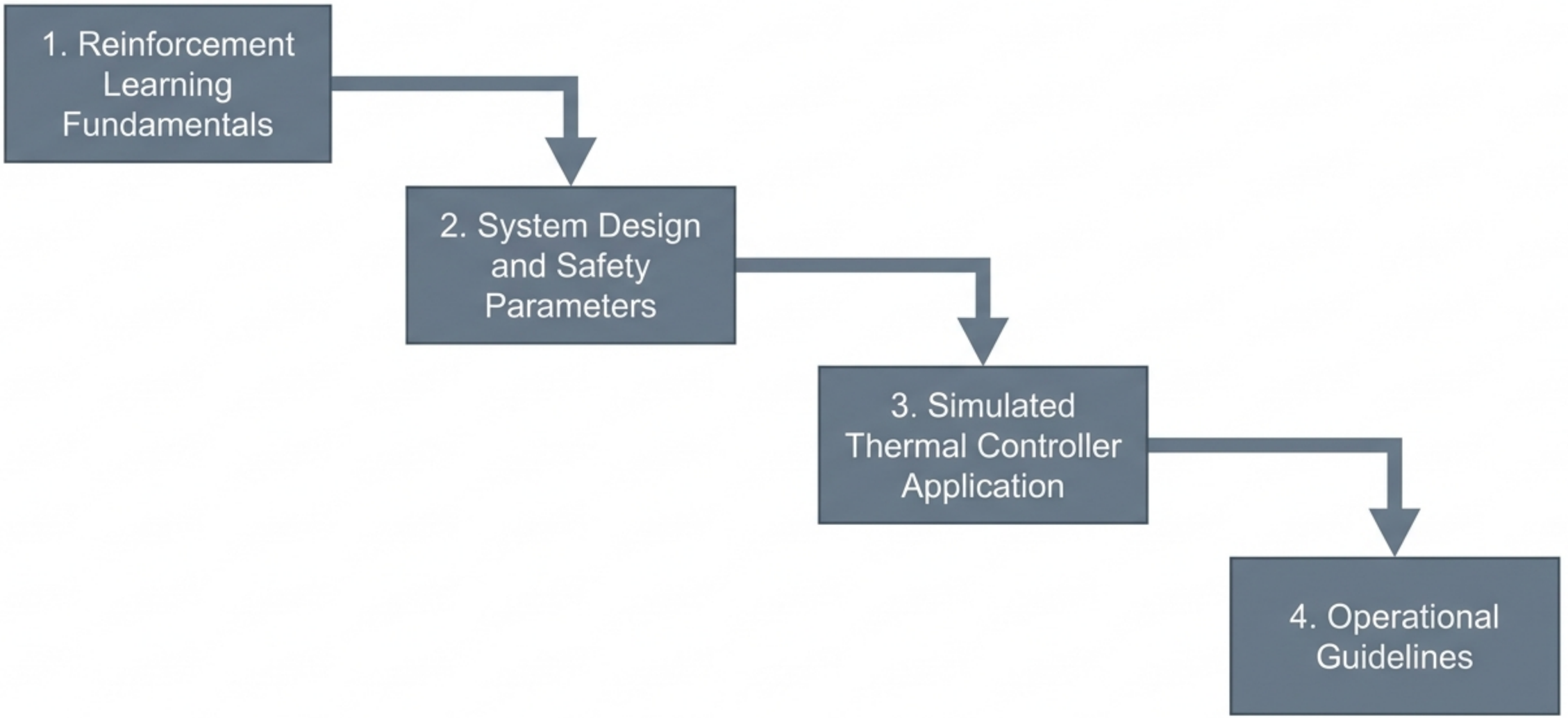


Module 4, Lesson 3
AI 101 for Mechanical Engineers
Reinforcement Learning Uses Interaction

Gemini Notebook

Structural Outline

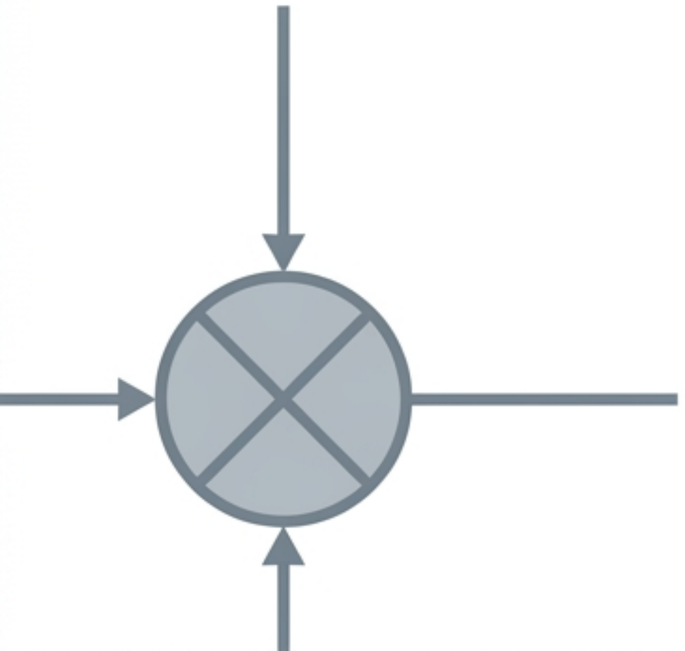


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2. Design & Safety
3. Thermal App
4. Guidelines

1. 1. Reinforcement Learning Fundamentals

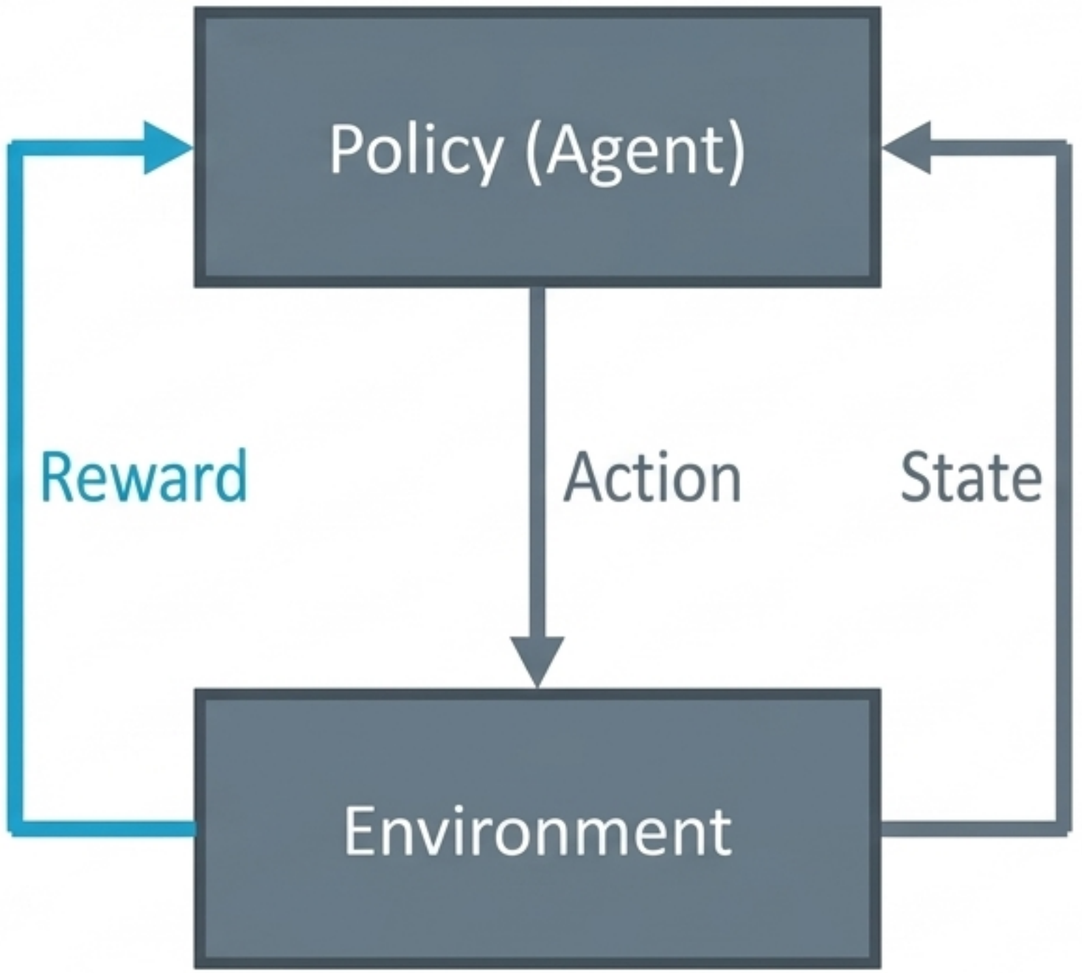
Define the core interactive variables and operational feedback loops governing autonomous control systems.



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Define The Interactive Control Loop

- Agent observes state information from the environment.
- Agent executes a selected control action.
- Environment returns interactive reward feedback.
- Policy systematically maps state data to actions.
- Objective seeks maximum cumulative reward over time.



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Distinguish Interactive From Supervised Learning

- Reinforcement learning requires active environment interaction.
- Supervised prediction evaluates static, pre-labeled datasets only.
- Labeled input-output examples lack dynamic interactive feedback.
- Interactive policies evaluate long-term action consequences.

	Reinforcement Learning	Supervised Prediction
Data Source	Interactive Environment	Labeled Examples
Feedback Type	Dynamic Rewards	Static Targets

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2. System Design and Safety Parameters

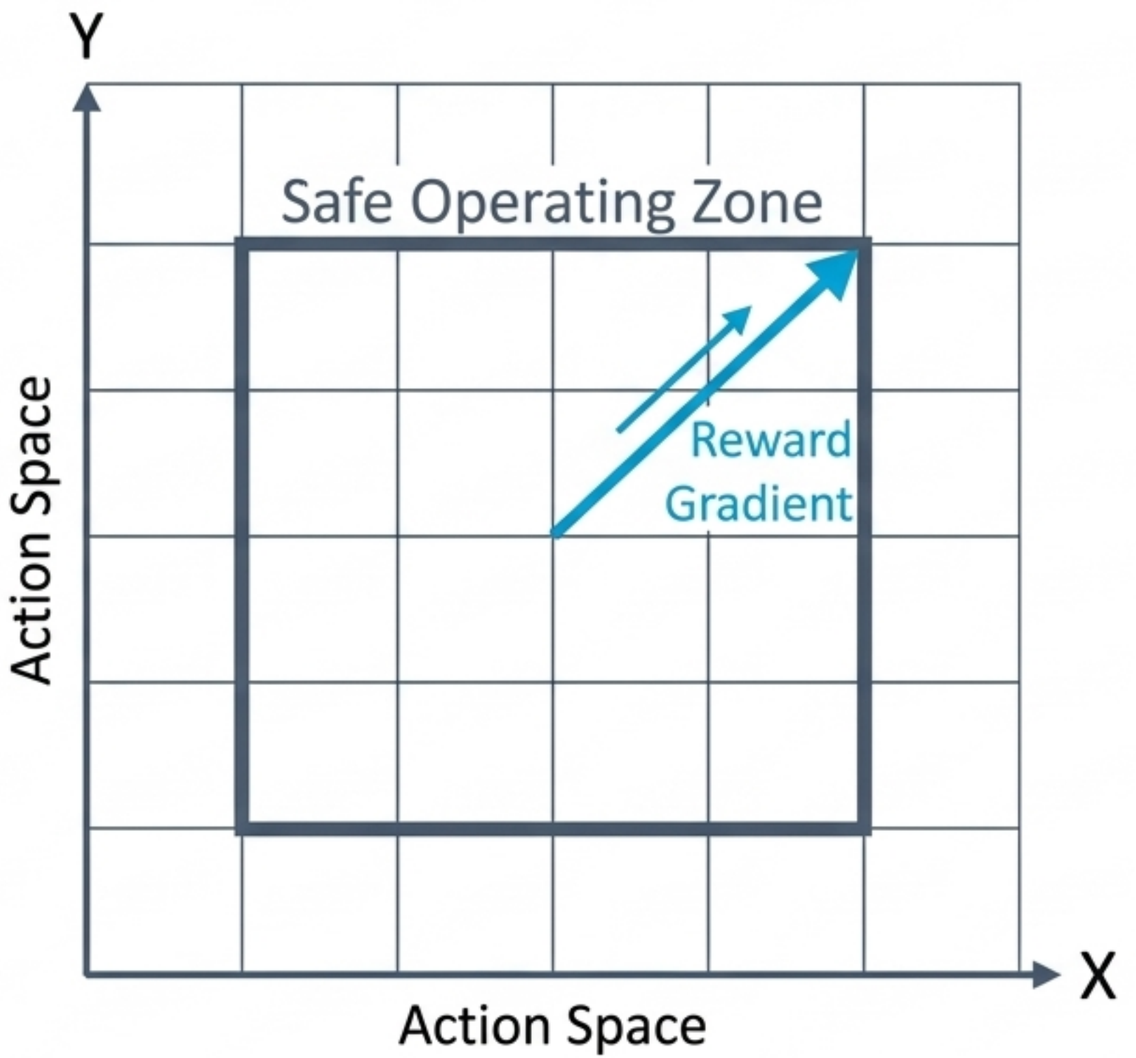
Establish strict boundaries between optimization objectives and non-negotiable operational safety limits.



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Separate Objectives From Operational Constraints

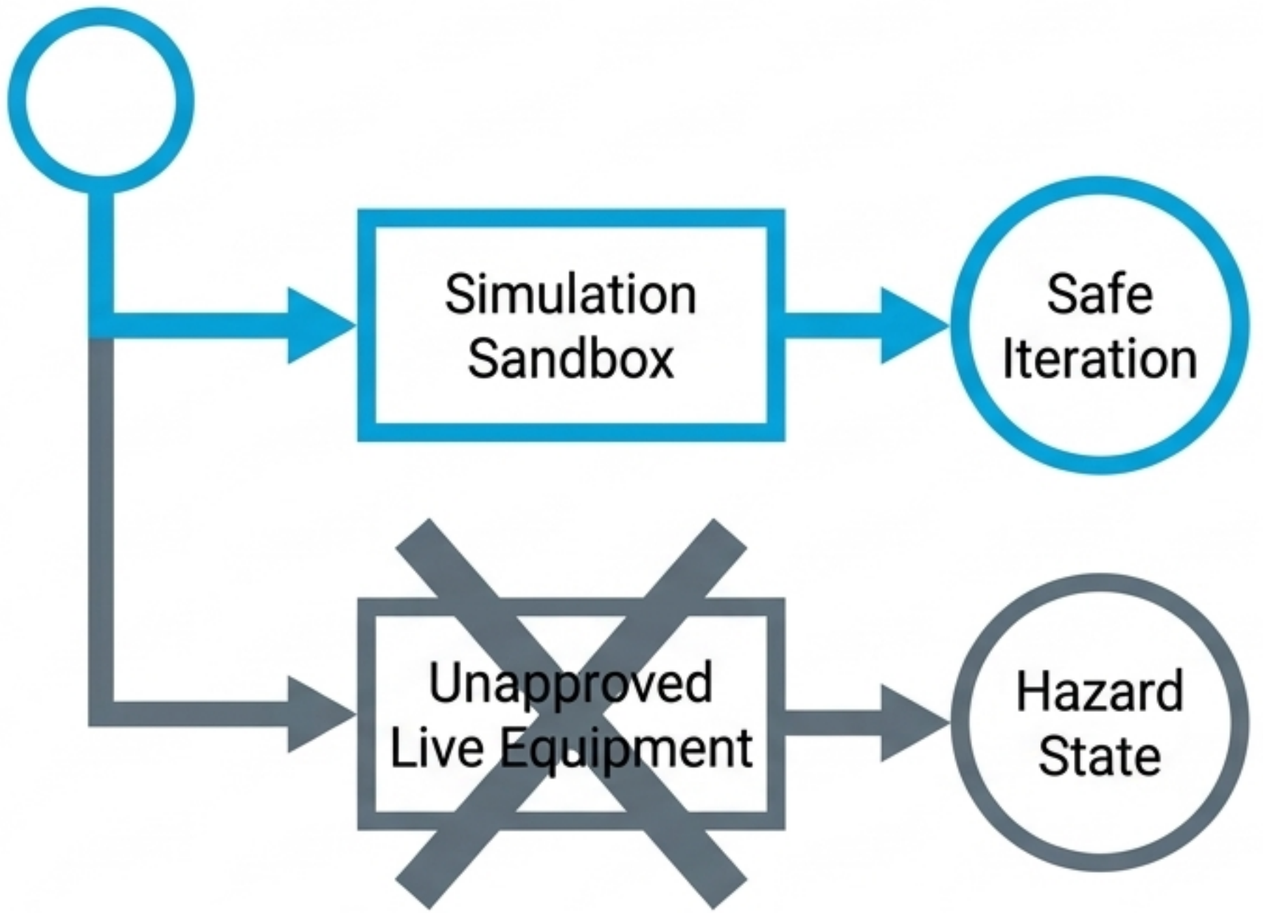
- Rewards strictly quantify the intended system objective.
- Poor rewards optimize numbers while violating purpose.
- Safety limits establish boundaries on allowable actions.
- Example: Battery controllers balance energy against degradation.



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Require Bounded Simulation For Exploration

- Beginner exercises strictly require simulated software environments.
- Live machinery experimentation demands explicit formal authorization.
- Interactive learning lacks inherent safe exploration guarantees.
- Physical testing requires established fallback mechanism validation.

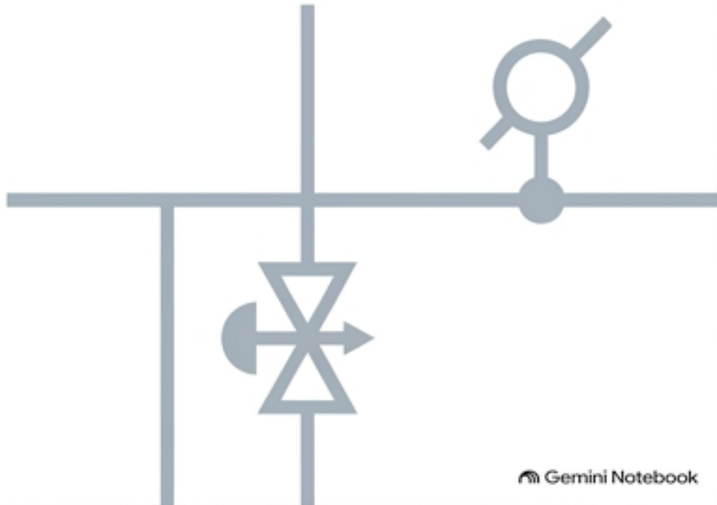


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3 3. Simulated Thermal Controller Application

Translate abstract interactive learning frameworks into standard mechanical control terminology.

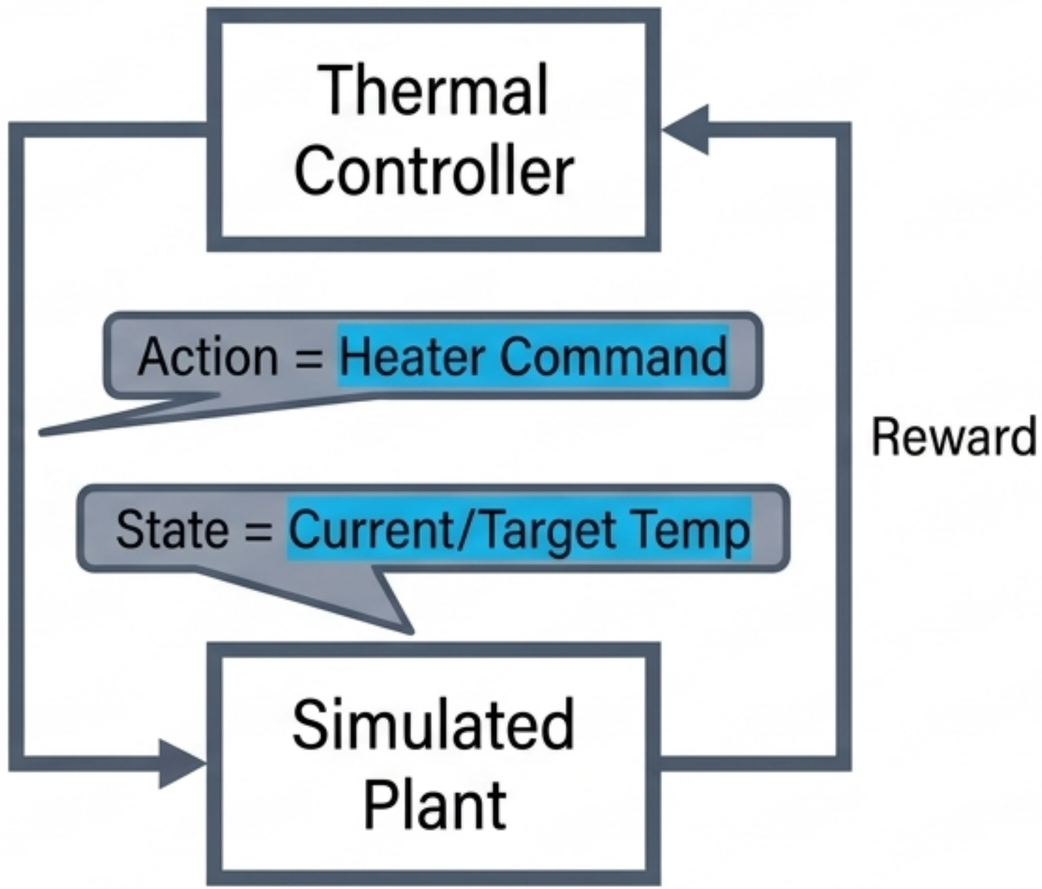


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Map Interactive Variables To Thermal States

- State comprises current and target system temperatures.
- Action dictates the specific hardware heater command.
- Policy executes rules mapping temperatures to commands.
- Simulation models the physical thermal plant response.



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Define Thermal Objectives And Safe Limits

- Reward continuously evaluates thermal tracking accuracy.
- Energy expenditure applies a calculated negative reward.
- Constraints define the absolute allowed temperature range.
- Safety constraints operate independently of reward functions.



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4

4. Operational Guidelines

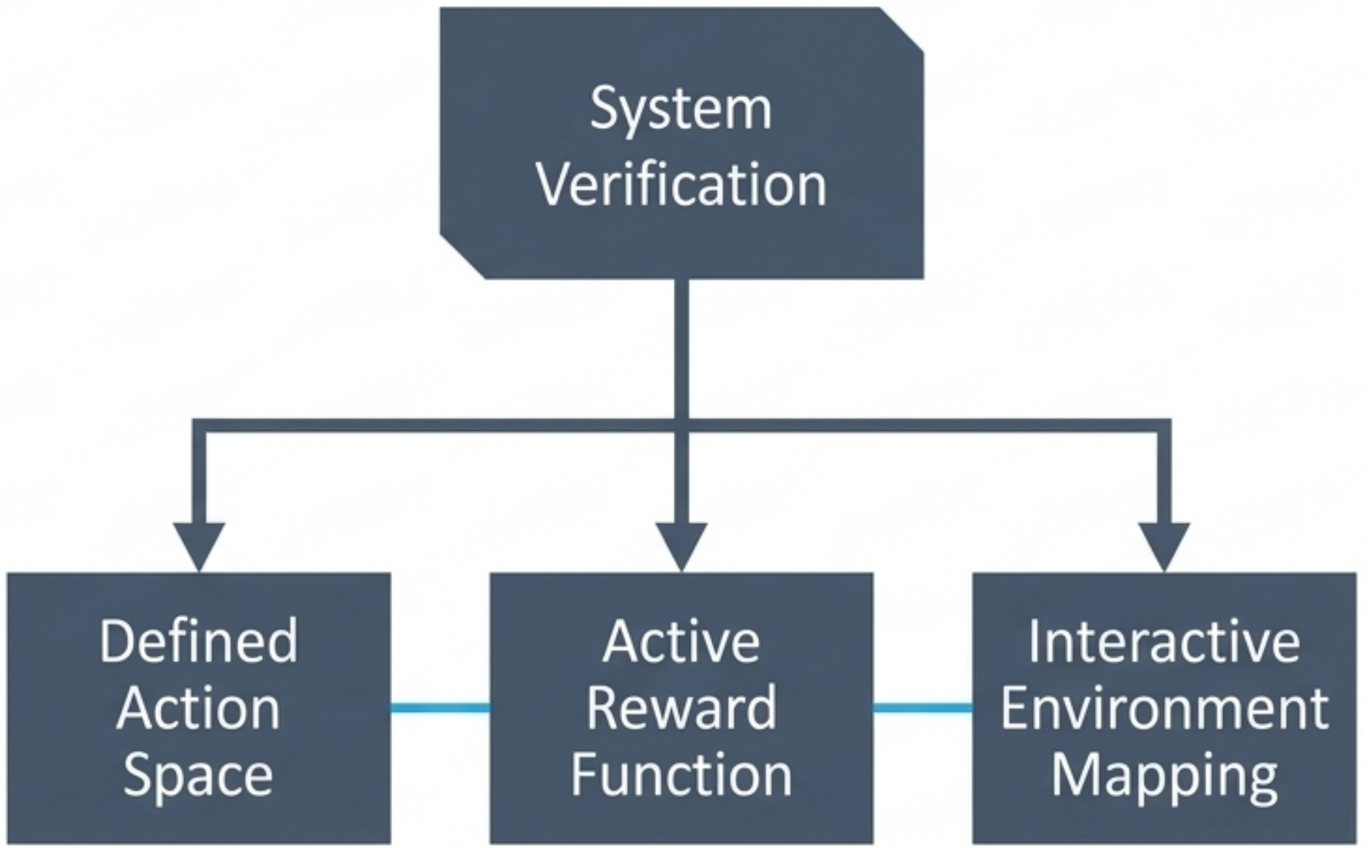
Consolidate system requirements and safety protocols for proper interactive control deployment.



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Verify Minimum System Setup Requirements

- Setup strictly demands defined actions and rewards.
- Setup strictly demands an interactive target environment.
- Policies must compute choices using available information.
- Safety verification requires distinct evidence and constraints.



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Key Points

1. Interactive systems operate via continuous state-action-reward loops.
2. Robust design strictly separates operational constraints from rewards.
3. Physical parameters translate directly into definable control states.
4. Initial interactive exploration strictly demands secure simulation environments.

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